

BINF 5503

Data Analytics and Storytelling

Week 1

Module 1 – Introduction to Data Analytics and Storytelling,
Project Details

Welcome!

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Ice Breaker Question



Please scan the QR Code for the Menti poll

Lecture overview

- What is storytelling?
- Tools: R, Jupyter Notebook on CoCalc
- Assessments
- Expectations of the course and classroom
- Datasets
- Research Questions

What is data storytelling?

- Communicating the story that data and the associated analysis tells, in a clear, effective manner
 - Uses data, narrative, and visualizations
- This is becoming an increasingly important skill that employers require across many industries

Why do we share data via storytelling?

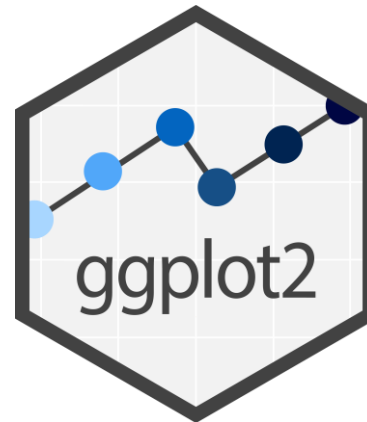
- Our brains take in so much information every day – what is important to process and remember?
- With effective storytelling, we can engage our audience in a way so that important information is retained
- We can also communicate the need for action

Course Structure

Weeks	Module	Topic
1	1	Introduction to Storytelling
2, 3	2	Data Analysis Methods
4	3	Best Practices for Data Visualization
5, 6	4	The Scientific Process
7	5	Reproducibility
Reading Week		
9	6	Shiny Apps
10, 11	7	Project Work Sessions & Presentations

Tools: R

- R is a rich statistical language with extensive bioinformatics support
- We will be using R for data analysis and visualizations
- Please follow the tidyverse style guide when coding:
<https://style.tidyverse.org/>



Tools: Jupyter Notebooks



Web-based interactive computing platform



Supports both R and Python



Notebooks will be running and submitted through cocalc.com, information to follow

Course Assessments

Assessment	Weight
Coding Assignments	24% (2 x 12%)
Worksheets	36% (3 x 12%)
Poster Presentation	16%
Final Poster	16%
In-Class Practice Problems	8% (8 x 1%)
Total	100%

Course Assessments – In-Class Practice

- 8 in total, 1% each, due 5:05 PM the day after class (Thursday)
 - Submit through Blackboard or CoCalc depending on the week
- For attendance & participation
- Concepts build on top of each other, so it is important to keep up with the practice!

Course Assessments – Worksheets

- 3 in total, 12% each, due on Friday at 8 PM of the given week
 - Will be released at least 2 weeks in advance of due date
 - Submit on Blackboard
- Aimed at soft skills: reading, writing, and data interpretation
- Everyone will receive feedback
- Read the questions carefully, review the rubric before you get started

Course Assessments – Jupyter Coding Assignments

- 2 in total, 12% each, due on Friday at 8 PM of the given week
 - Will be released at least 2 weeks in advance of due date
 - Submit as Jupyter Notebook through CoCalc
- No answer keys will be posted, there are multiple correct ways to solve the questions
- The “best” solutions will be written with efficient and readable code
- Everyone will receive feedback
- Read the questions carefully, review the rubric before you get started

Course Assessments – Final Presentation

More information on these assessments will be posted later in the semester!

- 5 datasets, groups of 2-4 students
- Analyze the data to create a fully-formed story using a poster
- At our BINF Conference, present on your findings and discuss with peers
- Each of the Jupyter assignments and worksheets will provide relevant tools for you to develop your project!
- Presentation (16%)
- Final Poster (16%)

Support

- Dedicated time at the beginning and end of each class for discussion, review, or answers to questions
- Post questions about course content first on the Blackboard Discussion Board in the relevant discussion category
 - Feel free to respond to your peers' questions (Marianne will review all posts)
 - Do not post completed code in your question or answer
- There will also be a help session on the 2nd last week of class for your presentation assignments
- Sensitive or personal questions should be directed to Marianne (Marianne.Toste@humber.ca) with "BINF5503" in the subject line

Code of Student Community Standards

<https://humber.ca/knowthecode/codes-community-standards>

1.1 Our Community Rights & Responsibility

All community members have a responsibility to conduct themselves in a manner that is consistent with the core values embraced by the College community and reflected in its various codes and policies.

AI Guidelines

- Under no circumstances should artificial intelligence be the primary source for final submissions for evaluations
- Acceptable uses of AI as a minor aid:
 - Help rewording a research question
 - As a starting point for finding resources on your topic
 - Proofreading - checking grammar and spelling
- Most obtain explicit permission for any aids not listed above
- Include an attribution statement for any AI assistance used

Missed and Late Evaluation Policy

- Ideally 24 hours before evaluation is due:
 - Extension form (<https://forms.office.com/r/8tD3Xtvms9>)
 - Email Marianne (Marianne.Toste@humber.ca)
- No-questions-asked 48-hour extension can be applied to any of the Jupyter Coding Assignments or Worksheets upon submission of the extension form
- 5% per day late penalty, up to 5 days otherwise

Project Datasets

1. Cancer Transcriptomics
2. Global Health
3. Multi-species Transcriptomics
4. Ecology
5. Social Sciences

1. Cancer Transcriptomics

Ovarian cancer in culture

- Malignant effusion specimens were collected from patients with high grade serous ovarian carcinoma (HGSOC)
- Data: RNA read-count data of four samples of multicellular spheroids grown in culture conditions taken at D0 and D6.
- Considerations:
 - Must have some prior knowledge or familiarity in cancer and drug testing
 - This is a simple dataset so you need to get creative!

2. Global Health

Pollution-Induced Migration

- Focuses on how air pollution in the city of Hanoi, Vietnam - one of the most polluted capital cities in the world - affects migration patterns of its workforce
- Data involves responses from 475 urban inhabitants from different areas
- Considerations:
 - This is a lot of data (66 columns) and there are many initial analysis that can be done to discover patterns
 - The data is in a binary format but does come with a key as to what each value means
 - Missing values are included – you must discuss how to handle them with your group

3. Multi-species Transcriptomics Circadian System and Aging

- Data entails raw RNA read counts of 2 species: *Danio rerio*, and *Nothobranchius furzer* from different tissue types (brain, liver and skin) at different time points.
- Considerations:
 - 116 raw files!!
 - Depending on how you structure your question you may not need them all
 - Gene codes are different in different species, though they may be referring to the same gene
 - This will be a difficult dataset to manage so tread wisely

4. Ecology

Ungulate herbivory under rainfall uncertainty

- Data collected in 2008
- A series of size-selective large-herbivore exclosures across a natural rainfall gradient in semi-arid Kenyan savanna
- 3 electrically fenced herbivore-exclusion treatments and one unfenced control was applied to blocks of contiguous 1-ha plots
- Each block of treatments is replicated 3 times at 3 different sites along a 20-km rainfall gradient
- Considerations:
 - 13 different datasets!
 - Difficulty is in selecting which dataset(s) to use to answer your question

5. Social Sciences

Studying African Farmer-Led Irrigation (SAFI)

- Data collected between November 2016 and June 2017
- The survey covered such things as: household features (e.g., construction materials used, number of household members), agricultural practices (e.g., water usage), assets (e.g., number and types of livestock) and details about the household members.
- Considerations:
 - Very wide data: lots of columns
 - Survey data & qualitative data comes with some pitfalls

Effects of age and chromosome type on density of genetic elements in *Rumex hastatulus*

01. Introduction

- Long Terminal Repeats retrotransposons (LTR-RTs) are genetic elements that can copy itself and move throughout the genome, thanks to the LTRs at its ends.⁹
- LTR-RTs often constitute 40% to 80% of plant genomes.^{4,9}
- One approach to investigating LTR-RTs involves analyzing their age, allowing for differentiation of older and younger LTRs, shedding light on their evolutionary history and activity.⁴
- Sex chromosomes often exhibit distinct genomic characteristics, including differences in gene content, repetitive element distribution, and evolutionary processes, which can influence LTR density.²
- The dataset utilized for our analysis pertains to the plant *Rumex hastatulus*, offering a unique opportunity to explore the genomic landscape of LTR-RTs within this species.



Figure 1. Flowering plant *Rumex hastatulus*

02. Research Questions

- Is there an effect of the evolutionary age of LTRs on both gene density and on LTR density in plant genomes?
- Is there a difference in LTR density between sex chromosomes and autosomal chromosomes?

Hypotheses:

- Question 1:** Older LTRs are more likely to reside in region with higher gene density and higher LTR density regions. Older LTRs accumulate over evolutionary time in regions less prone to mutation, and inversely younger LTRs should appear in regions of lower gene density and lower LTR density.
- Question 2:** Sex chromosomes will have higher LTR densities due to their lower recombination rates compared to autosomal chromosomes, therefore we think there will be more movement/proliferation of LTRs in these chromosomes.

03. Methodology

Data Description

- Set of highly maintained genes and a particular type of transposon found in a plant species called *Rumex hastatulus*.
- The data was provided by Vicki Zhang and Yeshoda-Harry Paul, who instruct the Data Analytics and Storytelling course at Humber College.
- Focus of analysis:** gene density, LTR density, LTR age, and chromosomes from the dataset.

Statistical Analysis

- Data analysis using R.
- Data exploration + Wrangling: base R and Tidyverse package
- The ggplot2 package was used for data visualization.
- The viridis package was used to make the figure colour-blind friendly.

LTR age on gene density and on LTR density:

- Base R linear modeling**

Differences in LTR density based on chromosome:

- was examined using **Kruskal-Wallis** sum rank test and **Dunn** test found in 'rstatix' package.

04. Results

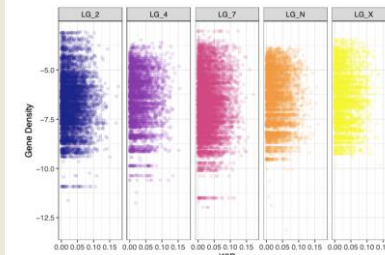


Figure 2. Effect of age of long-term repeats (LTRs) on gene density of *Rumex hastatulus*. The graph shows age of LTRs, depicted as K2P, on the x-axis and its effects on the gene density for each chromosome on the y-axis. Each chromosome is represented by a different colour. According to this graph, as age increases there is less dense clustering of gene density indicating some effect of LTR age, however the effect is non-significant ($p = 0.69$).

There is a reduction of clustering in gene density with increasing LTR age of *Rumex hastatulus* as depicted in figure 2, indicating some effect of LTR age on gene density. However, the linear model showed that the effect was non-significant ($p = 0.69$).

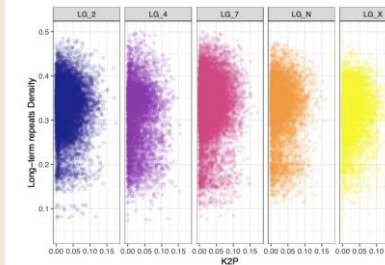


Figure 3. Effect of age of long-term repeats (LTRs) on LTR density of *Rumex hastatulus*. The graph shows age of LTRs, depicted as K2P, on the x-axis and its effects on the LTR density for each chromosome on the y-axis. Each chromosome is represented by a different colour. The graph indicates that as age increases there is less dense clustering of LTR density indicating some effect of LTR age, however the effect is non-significant ($p = 0.497$).

Moreover, there is a reduction of clustering in LTR density with increasing LTR age of *Rumex hastatulus* as illustrated in figure 3, indicating some effect of LTR age on LTR density. However, the linear model showed that the effect was non-significant ($p = 0.497$).

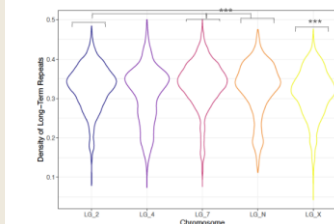


Figure 4. There is a difference between the long-term repeat (LTR) density of sex and autosomal chromosomes of *Rumex hastatulus*. The graph shows the LTR density on the y-axis for each chromosome on the x-axis. According to the graph, there is a difference in where the densities cluster among the chromosomes indicating difference of LTR densities among the chromosomes ($p < 2.2e-16$, chi-squared = 1152, df = 4). The sex chromosome LG_X had the largest difference in LTR density with the autosomal chromosome LG_7 ($p_{adj} = 5.98e-168$).

There is a difference in the location where the LTR densities cluster for the sex chromosome and the autosomal chromosomes as shown in figure 4, indicating that there is a difference among their density. The Kruskal-Wallis test indicated significant difference between at least two chromosomes ($p < 2.2e-16$). According to the Dunn-test for pair-wise comparison, the sex chromosome LG_X had the largest difference in density with the autosomal chromosome LG_7 ($p_{adj} = 5.98e-168$).

05. Conclusion

- The non-significant effect of LTR age on gene and LTR density is not consistent with the hypothesis. This can mean that are other factors such as the size of the genome affecting the gene and LTR densities of *Rumex hastatulus*.⁶
- Smaller genome size in plant species *C. melo* and *M. charantia* have shown with lower LTR densities. Larger genome size in *G. pentaphyllum* and *T. anguina* have shown to have higher LTR densities.⁶
- Other factors apart from survival from deleterious mutations should be studied for effect.
- There were significant differences in LTR density between sex chromosome and autosomes, which is consistent with the second hypothesis.
- It may be related to differences in recombination rates, also known as genetic shuffling. Recombination breaks LTRs.^{1,7}
- Higher genetic recombination rates in species such as *Oryza sativa* is associated with lower LTR density.¹⁰
- Lower recombination rates in sex chromosomes. Recombination is a major determinant of sex chromosomes' structure.^{1,7}
- Sex chromosomes are safer for LTRs
- Also limited by our data, we did not get a gene list, no context of where our LTRs are apart from density.

References

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Contributions

Urooj: background, conclusion, code, methods, results
Tanuj: research questions, hypothesis, background, code
Hasan: introduction, research questions, hypothesis, conclusion

Acknowledgements

Thank you to our professors, Vicki and Yeshoda, for your support, advice, and expertise, as well as to Humber College in providing us with the education and resources to complete this project.

Up next for projects

- Next week: gather into groups of 2-4 and select a dataset
 - Report your group member names and chosen dataset in Week 2 Practice Problems
- Outline (Worksheet 1, 12%, due Friday Jan 31st at 8 PM)
 - Research question, hypotheses, predictions
 - What statistical tests are you planning on using?
 - What are some limitations and/or roadblocks you anticipate?

Writing a strong research question

- Focused: specific to a single problem
 - Ask: what is the problem to be solved?
- Researchable: can be answered using the dataset
 - Ask: what variables am I investigating?
- Justifiable: addresses a research gap and is interesting
 - Ask: who cares? Why?
- Novel: should not copy questions investigated by other workers, but instead aim to confirm, refute, or discover new aspects of established findings
 - Ask: what have others done? How is your question the same? Different?

Is this a strong research question?

- What effect does social media have on your mind?💬
- More focused: What effect does daily use of Twitter have on the attention span of 12- to 16-year-olds?
- What should political parties do about low voter turnout in the region?💬
- More feasible & researchable: What are the most effective communication strategies for increasing voter turnout among those under 30 living within city limits?
- Is there a significant relationship between ice cream sales and Mars rover landings?💬
- More justifiable: Is there a significant relationship between salary and mental well-being of employees at the local shoe company?

Writing a hypothesis

- States your predictions about what your research will find
 - Phrase it as a tentative answer to your research question that has not yet been tested
- Includes relevant variables
- Related to research question
 - You may need to write several hypotheses that address different aspects of your research question
- Supported by background research

Is this a hypothesis?

- What are the health benefits of eating an apple a day?
 - This is a research question
- There is a relationship between social media use and attention span in under-16-year-old students.
 - Yes
- There is no relationship between social media use and attention span in under-16-year-old students.
 - Yes
- Higher levels of social media use result in lower attention spans in under-16-year-old students.
 - This errs on the side of prediction

Writing a prediction

- The outcome you would observe if your hypothesis were correct
- Often written in the form of “if-then” statements
 - “if my hypothesis is true, and I were to do this test, then this is what I will observe.”
- Try not to repeat the hypothesis verbatim

Putting it all together!

- Research Question: Why are there fewer asparagus beetles when asparagus is grown next to marigolds?
- Hypothesis: Marigolds deter asparagus beetles.
- Prediction: If marigolds deter asparagus beetles, and we grow asparagus next to marigolds, then we should find fewer asparagus beetles when asparagus plants are planted with marigolds.

Putting it all together!

- Research Question: What are the mechanisms driving KRAS inhibitor resistance in pancreatic cancer organoids?
- Hypothesis: Pancreatic cancer organoids exploit the alternative YAP/TAZ pathway to develop KRAS inhibitor resistance.
- Prediction: If the YAP/TAZ pathway is exploited by pancreatic cancer organoids to overcome KRAS inhibitors, differential expression analysis of RNA sequencing of KRAS inhibitor resistant organoids should show increased expression of YAP/TAZ pathways compared to control organoids.

Unexpected results

- It's exciting to get results, whether they support or refute your hypothesis!
- Null results are exceedingly common, and it is still a result!
 - However, they aren't nearly as likely to get published as "positive" results, even though they should be
 - Some null results represent potentially important discoveries, and leaving null results unpublished can also result in other researchers conducting the same study, wasting time and resources
- If you trust your results, make sure to justify why you got these results, and what that means for the field!